12-TONE EQUAL TEMPERAMENT: A LEAN 4 PROOF

KEITH BRAITHWAITE

Abstract. We present a Lean 4 formalisation of the theorem that 12-tone equal temperament (12tet) is the unique tuning system satisfying a small set of musically-motivated axioms. Intervals are modelled as elements of $\mathbb{Z}/m\mathbb{Z}$. The main theorem states $m = 12$; a subsidiary theorem shows the major/minor interval class has exactly 8 members.

Contents

1. Introduction 2
Model 2
The predicate isPerfect 2
2. Axioms 2
2.1. A1 and A2: the tritone 2
2.3. A4: perfect intervals are not self-inverse 3
2.4. A5: the perfect class is exactly $\{p, p^{-1}\}$ 3
2.5. A6: minimality of the perfect interval above the tritone 3
2.6. A7: five semitones above the perfect interval reaches unison 3
2.7. A8: a major/minor interval exists 4
3. Derived Facts 4
Shared consequences 4
m is even 4
$m \neq 2$ 5
$p = t + 1$, derived from A6 5
$m \mid 12$ 6
$m \neq 6$ and $m \neq 4$ 6
The 12TET theorem: $m = 12$ 7
Major/minor intervals as a least fixed point 7
Cardinality 10
4. The Partition Theorem 11
5. Risks to Validity 12
Axiom consistency 12
The numeral 5 in A7 13
The minimality axiom A6 13
The opaque predicate isPerfect 13
6. Further Work 13

Date: April 3, 2026.

1. Introduction

In 12-tone equal temperament (12TET) the octave is divided into 12 equal semitones. This paper asks: is there a small, musically well-motivated set of axioms that *forces* $m = 12$? We answer yes, by formalising the proof in Lean 4 with the Mathlib library.

Model. Harmonic intervals are modelled as elements of $\mathbb{Z}/m\mathbb{Z}$ for an unknown m. The four interval classes are:

Unison: the identity element, 0.
Tritone: the unique non-zero self-inverse element t ($t + t = 0$).
Perfect: a primitive class constrained by axioms; exactly the pair $\{p, p^{-1}\}$.
Major/Minor: the least fixed point of a ± 1 step operator starting from $\{p, p^{-1}\}$, stopping at $\{0, t\}$.

Inversion is additive negation: $i^{-1} = -i$. This gives $i + i^{-1} = 0$ by the ring axioms, corresponding to the musical fact that an interval and its inverse span an octave.

The predicate isPerfect. The predicate isPerfect is declared `opaque` in Lean: it has no explicit definition and can only be manipulated via the axioms. This is essential. Defining it as $i \neq 0 \wedge \neg\, \mathsf{isSelfInverse}\, i$ would make every major/minor interval "perfect", collapsing the four classes to two.

Listing 1. `Basic.lean`: the opaque perfect-interval predicate

```
opaque isPerfect [NeZero m] : HarmonicInterval m → Prop
```

2. Axioms

We use eight axioms. All are stated as Lean `axiom` declarations, which are trusted without proof.

2.1. A1 and A2: the tritone.

A non-zero self-inverse interval exists and is unique. A1 forces m to be even; A2 rules out $m = 8$ (where both 2 and 6 are self-inverse).

Listing 2. `Axioms.lean`: A1 and A2

```
axiom tritone_exists [NeZero m] :
    ∃ t : HarmonicInterval m, IsTritone t
axiom tritone_unique [NeZero m]
    (s t : HarmonicInterval m)
    (hs : IsTritone s)
    (ht : IsTritone t) : s = t
```

Listing 3. `Axioms.lean`: A3

```
axiom perfect_exists [NeZero m] : ∃ p : HarmonicInterval m, isPerfect p
```

2.3. **A4: perfect intervals are not self-inverse.** `perfect_nonzero` is derived immediately from A4: since unison is self-inverse, a perfect interval cannot be unison.

Listing 4. `Axioms.lean`: A4 and derived `perfect_nonzero`

```
axiom perfect_not_SI [NeZero m]
    (p : HarmonicInterval m) (hp : isPerfect p) : ¬isSelfInverse p
theorem perfect_nonzero [NeZero m]
    (p : HarmonicInterval m) (hp : isPerfect p) : p ≠ 0 :=
  fun h ⟹ perfect_not_SI p hp (h ▷ unison_is_self_inverse)
```

2.4. **A5: the perfect class is exactly $\{p, p^{-1}\}$.** `perfect_unique` and `perfect_inv_closed` are derived directly from A5 by taking the two directions of the biconditional.

Listing 5. `Axioms.lean`: A5 and derived consequences

```
axiom perfect_iff [NeZero m]
    (p : HarmonicInterval m) (hp : isPerfect p)
    (q : HarmonicInterval m) : isPerfect q ↔ q = p ∨ q = p⁻¹

theorem perfect_unique [NeZero m]
    (p q : HarmonicInterval m) (hp : isPerfect p) (hq : isPerfect q) : q = p ∨ q
    = p⁻¹ :=
  (perfect_iff p hp q).mp hq

theorem perfect_inv_closed [NeZero m]
    (p : HarmonicInterval m) (hp : isPerfect p) : isPerfect p⁻¹ :=
  (perfect_iff p hp p⁻¹).mpr (Or.inr rfl)
```

2.5. **A6: minimality of the perfect interval above the tritone.** Rather than asserting $p = t + 1$ directly (which would introduce an unexplained literal), A6 states that p is the *least* non-self-inverse element of $\mathbb{Z}/m\mathbb{Z}$ whose value exceeds that of t. The equation $p = t + 1$ is then *derived* in Section 3.

Listing 6. `Axioms.lean`: A6

```
axiom perfect_least_above_tritone [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    ZMod.val t < ZMod.val p ∧
    ∀ i : HarmonicInterval m, ¬isSelfInverse i →
      ZMod.val t < ZMod.val i → ZMod.val p ≤ ZMod.val i
```

2.6. **A7: five semitones above the perfect interval reaches unison.** This is the acoustic constraint. It encodes the approximation $\log_2(3/2) \approx 7/12$: the perfect fifth is 7 semitones, so its inverse (the perfect fourth) is 5 semitones, and $7 + 5 = 12 = 0$ in $\mathbb{Z}/12\mathbb{Z}$.

Listing 7. `Axioms.lean`: A7

```
axiom five_above_perfect [NeZero m]
    (p : HarmonicInterval m) (hp : isPerfect p) : p + 5 = 0
```

2.7. **A8: a major/minor interval exists.** This rules out $m = 4$. In $\mathbb{Z}/4\mathbb{Z}$, the four elements are fully accounted for by unison, tritone, and the perfect pair; there is no room for a fourth class.

Listing 8. `Axioms.lean`: A8

```
axiom major_minor_exists [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    ∃ i : HarmonicInterval m, i ≠ 0 ∧ ¬isSelfInverse i ∧ ¬isPerfect i
```

3. Derived Facts

Shared consequences. `tritone_eq` uses A2 (tritone uniqueness) to pin the tritone to a concrete value in any specific modulus, by supplying a witness and checking it satisfies the two defining properties. This eliminates repeated boilerplate in later proofs.

Listing 9. `Consequences.lean`: shared derived lemmas

```
theorem tritone_eq [NeZero m]
    (t : HarmonicInterval m) (ht : IsTritone t)
    (c : HarmonicInterval m) (hc0 : c ≠ 0) (hcsi : c + c = 0) : t = c :=
  tritone_unique t c ht { ne_zero := hc0, selfInverse := hcsi }

theorem selfInverse_eq_zero_or_tritone [NeZero m]
    (t i : HarmonicInterval m) (ht : IsTritone t) (hi : isSelfInverse i) :
    i = 0 ∨ i = t := by
  by_cases h : i = 0
  · exact Or.inl h
  · exact Or.inr (tritone_unique i t ⟨h, hi⟩ ht)

theorem perfect_not_self_inverse [NeZero m]
    (p : HarmonicInterval m) (hp : isPerfect p) : p ≠ p⁻¹ :=
  fun heq ⟹ perfect_not_SI p hp ((selfInverse_iff_inv_eq p).mpr heq.symm)


theorem perfect_pair_card [NeZero m]
    (p : HarmonicInterval m) (hp : isPerfect p) :
    ({p, p⁻¹} : Finset (HarmonicInterval m)).card = 2 :=
  Finset.card_pair (perfect_not_self_inverse p hp)
```

m is even. The tritone has additive order 2 (A1); the order of any element divides the group order $|\mathbb{Z}/m\mathbb{Z}| = m$; hence $2 \mid m$.

Listing 10. `TwelveTET.lean`: m is even

```
private lemma m_even [NeZero m]
    (t : HarmonicInterval m) (ht : IsTritone t) :
    2 ∣ m := by
  have hord : addOrderOf t = 2 := by
    haveI : Fact (Nat.Prime 2) := ⟨by decide⟩
    apply addOrderOf_eq_prime
    · rw [two_nsmul]; exact ht.selfInverse
    · exact ht.ne_zero
```

```
  have hdvd : addOrderOf t ∣ m := by
    have h : addOrderOf t ∣ Nat.card (ZMod m) := addOrderOf_dvd_natCard t
    rwa [Nat.card_eq_fintype_card, ZMod.card] at h  -- group order of ZMod m
      is m
  exact hord ▷ hdvd
```

$m \neq 2$**.** Every element of $\mathbb{Z}/2\mathbb{Z}$ is self-inverse, contradicting A4.

Listing 11. `TwelveTET.lean`: $m \neq 2$

```
private lemma m_ne_two [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    m ≠ 2 := by
  intro hm; subst hm
  exact perfect_not_SI p hp (show p + p = 0 from by fin_cases p <;> decide)
```

$p = t + 1$**, derived from A6.** Since m is even and $m \neq 2$ we have $m \geq 4$, so $2 \neq 0$ in $\mathbb{Z}/m\mathbb{Z}$. Therefore $t + 1$ is not self-inverse (its double is $2 \neq 0$). By A6, $\mathrm{val}(p) \leq \mathrm{val}(t + 1) = \mathrm{val}(t) + 1$. Combined with $\mathrm{val}(t) < \mathrm{val}(p)$ from A6, this forces $p = t + 1$.

Listing 12. `TwelveTET.lean`: $p = t + 1$ derived from minimality

```
lemma perfect_above_tritone [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    p = t + 1 := by
  obtain ⟨hgt, hmin⟩ := perfect_least_above_tritone p t hp ht
  have hm_pos : 0 < m := Nat.pos_of_ne_zero (NeZero.ne m)
  -- m ≥ 4: m is even and ≠ 2
  have hm4 : 4 ≤ m := by
    obtain ⟨k, hk⟩ := m_even t ht; have hn2 := m_ne_two p t hp ht; omega
  -- 2 ≠ 0 in ZMod m since m ≥ 4
  have h2ne : (2 : HarmonicInterval m) ≠ 0 := fun h ⟹
    absurd (Nat.le_of_dvd (by norm_num) ((CharP.cast_eq_zero_iff (ZMod m) m
    2).mp h))
           (by omega)
  -- t + 1 is not self-inverse: (t+1)+(t+1) = 2t+2 = 2 ≠ 0
  have ht1_nsi : ¬isSelfInverse (t + 1) := by
    intro h
    apply h2ne
    have : t + 1 + (t + 1) = 0 := h
    have : t + t = 0 := ht.selfInverse
    linear_combination ⟨t + 1 + (t + 1) = 0⟩ - ⟨t + t = 0⟩
  -- t + 1 ≠ 0
  have ht1_ne0 : t + 1 ≠ 0 := fun h ⟹ ht1_nsi (h ▷ unison_is_self_inverse
    )
  -- val(t + 1) = val t + 1: no wraparound since val t + 1 < m
  have hv1 : ZMod.val (1 : HarmonicInterval m) = 1 := ZMod.val_one'' (by omega
    )
  have hlt_m : ZMod.val t + ZMod.val (1 : HarmonicInterval m) < m := by
    rw [hv1]
    by_contra h
```

```
    push Not at h
    have heq : ZMod.val t + 1 = m := by have := ZMod.val_lt t; omega
    exact ht1_ne0 ((ZMod.val_eq_zero _).mp (by rw [ZMod.val_add, hv1, heq,
    Nat.mod_self]))
  have hval : ZMod.val (t + 1) = ZMod.val t + 1 := by
    rw [ZMod.val_add_of_lt hlt_m, hv1]
  -- By minimality: val p ≤ val(t+1) = val t + 1
  have hle : ZMod.val p ≤ ZMod.val (t + 1) :=
    hmin (t + 1) ht1_nsi (by omega)
  -- val p = val(t+1), so p = t+1
  exact ZMod.val_injective m (by omega)
```

$m \mid 12$**.** From $p = t + 1$ and A7: $t + 6 = (t + 1) + 5 = p + 5 = 0$. Then $12 = 2(t + 6) - (t + t) = 0$ in $\mathbb{Z}/m\mathbb{Z}$, so $m \mid 12$.

Listing 13. `TwelveTET.lean`: $m \mid 12$

```
private lemma tritone_plus_six [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    t + 6 = 0 :=
  calc t + 6 = t + 1 + 5 := by ring
    _        = p + 5     := by rw [← perfect_above_tritone p t hp ht]
    _        = 0         := five_above_perfect p hp
private lemma twelve_eq_zero [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    (12 : HarmonicInterval m) = 0 := by
  have hsi : t + t = 0 := ht.selfInverse
  linear_combination 2 * tritone_plus_six p t hp ht - hsi
private lemma m_dvd_twelve [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    m ∣ 12 :=
  (CharP.cast_eq_zero_iff (ZMod m) m 12).mp (twelve_eq_zero p t hp ht)
```

$m \neq 6$ **and** $m \neq 4$**.** In $\mathbb{Z}/6\mathbb{Z}$: $t = 3$, $p = 4$, $p + 5 = 9 \neq 0$, contradicting A7. In $\mathbb{Z}/4\mathbb{Z}$: $t = 2$, $p = 3$, $p^{-1} = 1$; every element is in $\{0, t, p, p^{-1}\}$, contradicting A8.

Listing 14. `TwelveTET.lean`: eliminating $m = 6$ and $m = 4$

```
private lemma m_ne_six [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    m ≠ 6 := by
  intro hm; subst hm
  have ht3 : t = 3 := tritone_eq t ht 3 (by decide) (by decide)
  have hp4 : p = 4 := by rw [perfect_above_tritone p t hp ht, ht3]; decide
  exact absurd (five_above_perfect p hp) (by simp only [hp4]; decide)
private lemma m_ne_four [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    m ≠ 4 := by
  intro hm; subst hm
  obtain ⟨i, hine, hiSI, hiP⟩ := major_minor_exists p t hp ht
  have ht2 : t = 2 := tritone_eq t ht 2 (by decide) (by decide)
  have hp3 : p = 3 := by rw [perfect_above_tritone p t hp ht, ht2]; decide
```

```
  have hpi : p⁻¹ = 1 := by simp only [HI_inv_eq_neg, hp3]; decide
  fin_cases i <;> simp only at *
  · exact hine rfl                                          -- i = 0: unison
  · exact hiP (by have := perfect_inv_closed p hp; rwa [hpi] at this)  -- i = 1 = p⁻¹
  · exact hiSI (by simp only [isSelfInverse]; decide)          -- i = 2 = t
  · exact hiP (by rwa [hp3] at hp)                              -- i = 3 = p
```

The 12TET theorem: $m = 12$**.** Since $m \mid 12$, m is even, and $m \notin \{2, 4, 6\}$, the only remaining possibilities are $m \in \{12\}$ (with $m = 1$ excluded by `NeZero`). The `interval_cases` tactic dispatches the finite case split.

Listing 15. `TwelveTET.lean`: $m = 12$

```
theorem twelve_TET [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    m = 12 := by
  have h1 := m_dvd_twelve p t hp ht
  have h2 := m_even t ht
  have h3 := m_ne_two p t hp ht
  have h4 := m_ne_four p t hp ht
  have h5 := m_ne_six p t hp ht
  have hm_pos : 0 < m := Nat.pos_of_ne_zero (NeZero.ne m)
  have hm_le : m ≤ 12 := Nat.le_of_dvd (by norm_num) h1
  interval_cases m <;> omega
```

Major/minor intervals as a least fixed point. The major/minor class is defined as the least fixed point of the step operator

$$\mathrm{stepMM}(p, t, s) = \big(s \cup \{j + 1 \mid j \in s \cup \{p, p^{-1}\}\} \cup \{j - 1 \mid j \in s \cup \{p, p^{-1}\}\}\big) \setminus \{0, t\}.$$

The lfp is the intersection of all prefixed points (sets s with $\mathrm{stepMM}(p, t, s) \subseteq s$).

Listing 16. `MajorMinor.lean`: definitions of selfInverseSet, stepMM, nonSIReachable, majorMinorIntervals

```
def selfInverseSet [NeZero m] (t : HarmonicInterval m) : Finset (HarmonicInterval m) :=
  {0, t}
def stepMM [NeZero m]
    (p t : HarmonicInterval m)
    (s : Finset (HarmonicInterval m)) : Finset (HarmonicInterval m) :=
  let frontier := s ∪ {p, p⁻¹}
  (s ∪ frontier.image (· + 1) ∪ frontier.image (· - 1)) \ selfInverseSet t
private theorem stepMM_monotone [NeZero m] (p t : HarmonicInterval m) :
    Monotone (stepMM p t) := by
  intro s s' hss' i hi
  simp only [stepMM, selfInverseSet, Finset.mem_sdiff, Finset.mem_union,
             Finset.mem_image, Finset.mem_insert, Finset.mem_singleton] at *
  obtain ⟨h, hni⟩ := hi
  refine ⟨?_, hni⟩
  rcases h with ((hs | hi1) | hi2)
```

```
  · exact Or.inl (Or.inl (hss' hs))
  · obtain ⟨j, hj, rfl⟩ := hi1
    rcases hj with hs | hp
    · exact Or.inl (Or.inr ⟨j, Or.inl (hss' hs), rfl⟩)
    · exact Or.inl (Or.inr ⟨j, Or.inr hp, rfl⟩)
  · obtain ⟨j, hj, rfl⟩ := hi2
    rcases hj with hs | hp
    · exact Or.inr ⟨j, Or.inl (hss' hs), rfl⟩
    · exact Or.inr ⟨j, Or.inr hp, rfl⟩
noncomputable def nonSIReachable [NeZero m]
    (p t : HarmonicInterval m) : Finset (HarmonicInterval m) :=
  Finset.univ.filter (fun i ⟹
    ∀ s : Finset (HarmonicInterval m), stepMM p t s ⊆ s → i ∈ s)

private theorem nonSIReachable_least [NeZero m]
    (p t : HarmonicInterval m) (s : Finset (HarmonicInterval m))
    (hs : stepMM p t s ⊆ s) : nonSIReachable p t ⊆ s := by
  intro i hi
  unfold nonSIReachable at hi
  rw [Finset.mem_filter] at hi
  exact hi.2 s hs
noncomputable def majorMinorIntervals [NeZero m]
    (p t : HarmonicInterval m) : Finset (HarmonicInterval m) :=
  nonSIReachable p t \ {p, p⁻¹}
```

The lfp equals the complement of the self-inverse set. The upper bound is immediate from monotonicity; the lower bound is proved by a 10-step walk in $\mathbb{Z}/12\mathbb{Z}$ starting from $p = 7$ and $p^{-1} = 5$.

Listing 17. `MajorMinor.lean`: upper bound

```
private theorem complement_SI_is_fixedPoint [NeZero m]
    (p t : HarmonicInterval m) :
    stepMM p t (Finset.univ \ selfInverseSet t) = Finset.univ \ selfInverseSet t
    := by
  ext i
  simp only [stepMM, selfInverseSet, Finset.mem_sdiff, Finset.mem_univ,
    true_and,
            Finset.mem_union, Finset.mem_image]
  constructor
  · rintro ⟨_, hni⟩; exact hni
  · intro hni
    -- i ∈ univ \ selfInverseSet t, so i ∈ s (the first component of the union
    )
    exact ⟨Or.inl (Or.inl hni), hni⟩
private theorem nonSIReachable_subset_complement [NeZero m]
    (p t : HarmonicInterval m) :
    nonSIReachable p t ⊆ Finset.univ \ selfInverseSet t :=
  nonSIReachable_least p t _ (complement_SI_is_fixedPoint p t).le
```

Listing 18. `MajorMinor.lean`: lower bound (10-step walk) and equality

```
private theorem prefixed_contains_complement [NeZero m]
    (p t : HarmonicInterval m)
    (hp : isPerfect p)
    (ht : IsTritone t)
    (s : Finset (HarmonicInterval m))
    (hpre : stepMM p t s ⊆ s) :
    Finset.univ \ selfInverseSet t ⊆ s := by
  have hm := twelve_TET p t hp ht
  subst hm
  have ht6 : t = 6 := tritone_eq t ht 6 (by decide) (by decide)
  have hp7 : p = 7 := by
    have := perfect_above_tritone p t hp ht; rw [ht6] at this; simpa using
    this
  have hpi5 : p⁻¹ = 5 := by
    simp only [HI_inv_eq_neg, hp7]; decide
  subst ht6; subst hp7
  -- {7, 5} = {p, p⁻¹};  {0, 6} = selfInverseSet t
  have step : ∀ j : ZMod 12,
      j ∈ s ∪ ({7, (5 : ZMod 12)} : Finset (ZMod 12)) →
      ∀ i : ZMod 12, (i = j + 1 ∨ i = j - 1) → i ∉ ({0, (6 : ZMod 12)} :
    Finset _) → i ∈ s := by
    intro j hj i hi hni
    apply hpre
    simp only [stepMM, selfInverseSet, Finset.mem_sdiff, Finset.mem_union,
               Finset.mem_image, hpi5]
    refine ⟨?_, hni⟩
    rcases hi with rfl | rfl
    · exact Or.inl (Or.inr ⟨j, Finset.mem_union.mp hj, rfl⟩)  -- i = j+1:
    frontier.image(·+1)
    · exact Or.inr ⟨j, Finset.mem_union.mp hj, rfl⟩            -- i = j-1:
    frontier.image(·-1)
  have fp7 : (7 : ZMod 12) ∈ s ∪ ({7, (5:ZMod 12)} : Finset _) := by simp
  have fp5 : (5 : ZMod 12) ∈ s ∪ ({7, (5:ZMod 12)} : Finset _) := by simp
  have h1 : (8 : ZMod 12) ∈ s :=
    step 7 fp7 8 (Or.inl (by decide)) (by decide)
  have h2 : (9 : ZMod 12) ∈ s :=
    step 8 (Finset.mem_union_left _ h1) 9 (Or.inl (by decide)) (by decide)
  have h3 : (10 : ZMod 12) ∈ s :=
    step 9 (Finset.mem_union_left _ h2) 10 (Or.inl (by decide)) (by decide)
  have h4 : (11 : ZMod 12) ∈ s :=
    step 10 (Finset.mem_union_left _ h3) 11 (Or.inl (by decide)) (by decide)
  have h5 : (4 : ZMod 12) ∈ s :=
    step 5 fp5 4 (Or.inr (by decide)) (by decide)
  have h6 : (3 : ZMod 12) ∈ s :=
    step 4 (Finset.mem_union_left _ h5) 3 (Or.inr (by decide)) (by decide)
  have h7 : (2 : ZMod 12) ∈ s :=
    step 3 (Finset.mem_union_left _ h6) 2 (Or.inr (by decide)) (by decide)
  have h8 : (1 : ZMod 12) ∈ s :=
```

```
      step 2 (Finset.mem_union_left _ h7) 1 (Or.inr (by decide)) (by decide)
  have h9 : (7 : ZMod 12) ∈ s :=
      step 8 (Finset.mem_union_left _ h1) 7 (Or.inr (by decide)) (by decide)
  have h10 : (5 : ZMod 12) ∈ s :=
      step 4 (Finset.mem_union_left _ h5) 5 (Or.inl (by decide)) (by decide)
  intro i hi
  simp only [selfInverseSet, Finset.mem_sdiff, Finset.mem_univ, true_and,
             Finset.mem_insert, Finset.mem_singleton] at hi
  push Not at hi
  obtain ⟨hi0, hi6⟩ := hi
  fin_cases i
  · exact absurd rfl hi0
  · exact h8
  · exact h7
  · exact h6
  · exact h5
  · exact h10
  · exact absurd rfl hi6
  · exact h9
  · exact h1
  · exact h2
  · exact h3
  · exact h4
private theorem complement_subset_nonSIReachable [NeZero m]
    (p t : HarmonicInterval m)
    (hp : isPerfect p)
    (ht : IsTritone t) :
    Finset.univ \ selfInverseSet t ⊆ nonSIReachable p t := by
  intro i hi
  unfold nonSIReachable
  rw [Finset.mem_filter]
  exact ⟨Finset.mem_univ _, fun s hs ⟹ prefixed_contains_complement p t
    hp ht s hs hi⟩
theorem nonSIReachable_eq_complement [NeZero m]
    (p t : HarmonicInterval m)
    (hp : isPerfect p)
    (ht : IsTritone t) :
    nonSIReachable p t = Finset.univ \ selfInverseSet t :=
  Finset.Subset.antisymm
    (nonSIReachable_subset_complement p t)
    (complement_subset_nonSIReachable p t hp ht)
```

Cardinality.

Theorem 3.1. *The major/minor class has $m - 4$ elements; in* 12TET*, exactly 8.*

Listing 19. `Cardinality.lean`

```
theorem selfInverseSet_card [NeZero m]
    (t : HarmonicInterval m) (ht : IsTritone t) :
```

```
    ( selfInverseSet  t).card = 2 :=
  Finset.card_pair (Ne.symm ht.ne_zero)

private theorem nonSIReachable_card [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    (nonSIReachable p t).card = m - 2 := by
  rw [nonSIReachable_eq_complement p t hp ht, Finset.card_sdiff]
  simp [Finset.card_univ, ZMod.card, selfInverseSet_card t ht]

private theorem perfect_pair_subset_nonSIReachable [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    ({p, p⁻¹} : Finset (HarmonicInterval m)) ⊆ nonSIReachable p t := by
  rw [nonSIReachable_eq_complement p t hp ht]
  simp only [selfInverseSet, Finset.subset_sdiff, Finset.subset_univ, true_and,

             Finset.disjoint_left, Finset.mem_insert, Finset.mem_singleton]
  intro i hi
  rcases hi with rfl | rfl
  · simp only [not_or]
    exact ⟨perfect_nonzero _ hp, fun h ⟹ perfect_not_SI _ hp (h ▷ ht.
    selfInverse)⟩
  · simp only [not_or]
    exact ⟨by simp [HI_inv_eq_neg, neg_eq_zero, perfect_nonzero _ hp],
           fun h ⟹ perfect_not_SI _ (perfect_inv_closed _ hp) (h ▷ ht.
    selfInverse)⟩
theorem majorMinor_card [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    (majorMinorIntervals p t).card = m - 4 := by
  simp only [majorMinorIntervals]
  have hsub := perfect_pair_subset_nonSIReachable p t hp ht
  have hcard_r := nonSIReachable_card p t hp ht
  have hcard_p := perfect_pair_card p hp
  rw [Finset.card_sdiff]
  have hint : {p, p⁻¹} ∩ nonSIReachable p t = {p, p⁻¹} :=
    Finset.inter_eq_left.mpr hsub
  rw [hint, hcard_r, hcard_p]; omega
theorem majorMinor_card_12 [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
    (majorMinorIntervals p t).card = 8 := by
  rw [majorMinor_card p t hp ht, twelve_TET p t hp ht]
```

4. The Partition Theorem

Theorem 4.1. *The four classes — $\{0\}$, $\{t\}$, $\{p, p^{-1}\}$, and the major/minor class — partition $\mathbb{Z}/m\mathbb{Z}$.*

Listing 20. `Partition.lean`

```
theorem interval_partition [NeZero m]
    (p t : HarmonicInterval m) (hp : isPerfect p) (ht : IsTritone t) :
```

```
    (Finset.univ : Finset (HarmonicInterval m)) =
      selfInverseSet t ∪ {p, p⁻¹} ∪ majorMinorIntervals p t := by
  simp only [majorMinorIntervals, selfInverseSet]
  rw [nonSIReachable_eq_complement p t hp ht]
  ext i
  simp only [Finset.mem_univ, Finset.mem_union, Finset.mem_insert, Finset.
    mem_singleton,
             Finset.mem_sdiff, true_iff]
  by_cases h0  : i = 0;    · exact Or.inl (Or.inl (Or.inl h0))
  by_cases ht' : i = t;    · exact Or.inl (Or.inl (Or.inr ht'))
  by_cases hp' : i = p;    · exact Or.inl (Or.inr (Or.inl hp'))
  by_cases hpi : i = p⁻¹; · exact Or.inl (Or.inr (Or.inr hpi))
  · right
    refine ⟨⟨trivial, ?_⟩, ?_⟩
    · simp only [selfInverseSet, Finset.mem_insert, Finset.mem_singleton,
    not_or]
      exact ⟨h0, ht'⟩
    · simp only [not_or]
      exact ⟨hp', by rwa [HI_inv_eq_neg]⟩
```

5. Risks to Validity

Axiom consistency. Since isPerfect is opaque, the axioms cannot be proved as theorems. We address this by bundling the eight axiom conditions into a structure `HarmonicModel`, parameterised over an explicit predicate `perf` that stands in for isPerfect. A term of this type for a specific modulus is a consistency certificate.

Listing 21. `Model.lean`: axiom bundle and consistency certificate

```
structure HarmonicModel (m : ℕ) [NeZero m] where
  t    : HarmonicInterval m
  p    : HarmonicInterval m
  perf : HarmonicInterval m → Prop
  A1   : IsTritone t
  /-- A2: the tritone is unique. -/
  A2   : ∀ s : HarmonicInterval m, IsTritone s → s = t
  /-- A3: a perfect interval exists. -/
  A3   : perf p
  /-- A4: perfect intervals are not self-inverse. -/
  A4   : ∀ q, perf q → ¬isSelfInverse q
  /-- A5: the perfect class is exactly {p, p⁻¹}. -/
  A5   : ∀ q, perf q ↔ q = p ∨ q = p⁻¹
  /-- A6: p is the least non-self-inverse interval above t. -/
  A6   : ZMod.val t < ZMod.val p ∧
         ∀ i : HarmonicInterval m, ¬isSelfInverse i →
           ZMod.val t < ZMod.val i → ZMod.val p ≤ ZMod.val i
  /-- A7: five semitones above p reaches unison. -/
  A7   : p + 5 = 0
  /-- A8: a non-unison non-self-inverse non-perfect interval exists. -/
  A8   : ∃ i : HarmonicInterval m, i ≠ 0 ∧ ¬isSelfInverse i ∧ ¬perf i
```

```
/-- The standard model: `ZMod 12`, `t = 6`, `p = 7`, `perf = {7, 5}`.
All conditions are verified by `decide` or `fin_cases` + `decide`. -/
def zmod12Model : HarmonicModel 12 where
  t    := 6
  p    := 7
  perf := fun i ⟹ i = 7 ∨ i = 5
  A1   := ⟨by decide, by decide⟩
  A2   := by
    intro s ⟨hne, hsi⟩
    fin_cases s <;> simp_all [isSelfInverse]
  A3   := Or.inl rfl
  A4   := by
    intro q hq
    rcases hq with rfl | rfl <;> simp only [isSelfInverse] <;> decide
  A5   := by
    intro q
    simp only [HI_inv_eq_neg]
    fin_cases q <;> decide
  A6   := ⟨by decide, by
    intro i hsi hgt
    fin_cases i <;> revert hsi hgt <;> simp only [isSelfInverse] <;> decide⟩
  A7   := by decide
  A8   := ⟨1, by decide, by simp only [isSelfInverse]; decide, by decide⟩
```

The model is $\mathbb{Z}/12\mathbb{Z}$ with $t = 6$, $p = 7$, and $\mathrm{perf}\, i \Leftrightarrow i = 7 \vee i = 5$. All eight conditions are discharged computationally by `decide` or `fin_cases` followed by `decide`.

The numeral 5 in A7. Axiom A7 ($p + 5 = 0$) introduces the literal 5. This is the irreducible acoustic content of the proof: the perfect fifth is approximated by 7/12 of an octave, a fact about physics that cannot be derived from qualitative musical structure alone. Any reformulation of A7 that avoids the numeral 5 must implicitly encode the same information.

The minimality axiom A6. The minimality formulation of A6 avoids the literal 1 of the original $p = t + 1$, but the derived lemma `perfect_above_tritone` still concludes $p = t + 1$. The improvement is that the *axiom* is now qualitative; the numeral emerges from the proof rather than being assumed.

The opaque predicate isPerfect. Declaring isPerfect opaque is load-bearing: it prevents Lean from unfolding a hypothetical explicit definition during proof search, which could produce spurious proofs. The trade-off is that isPerfect cannot be computationally evaluated.

6. Further Work

- **Formalise a model.** Construct $\mathbb{Z}/12\mathbb{Z}$ explicitly with $t = 6$, $p = 7$, and verify all eight axioms, giving a consistency proof.
- **Eliminate A7.** Replace $p + 5 = 0$ with an axiom expressing acoustic nearness (best rational approximation of $\log_2(3/2)$) and derive $p + 5 = 0$

as a theorem. This requires bringing in real analysis or continued-fraction arithmetic.

- **Diatonic structure.** Characterise the major scale as a specific 7-element subset of the major/minor class and derive its interval structure (tone–semitone pattern) from the axioms.
- **Voice-leading.** Formalise the claim that the major/minor class is closed under minimal voice-leading displacement (± 1 step), connecting the lfp construction to the musical concept.